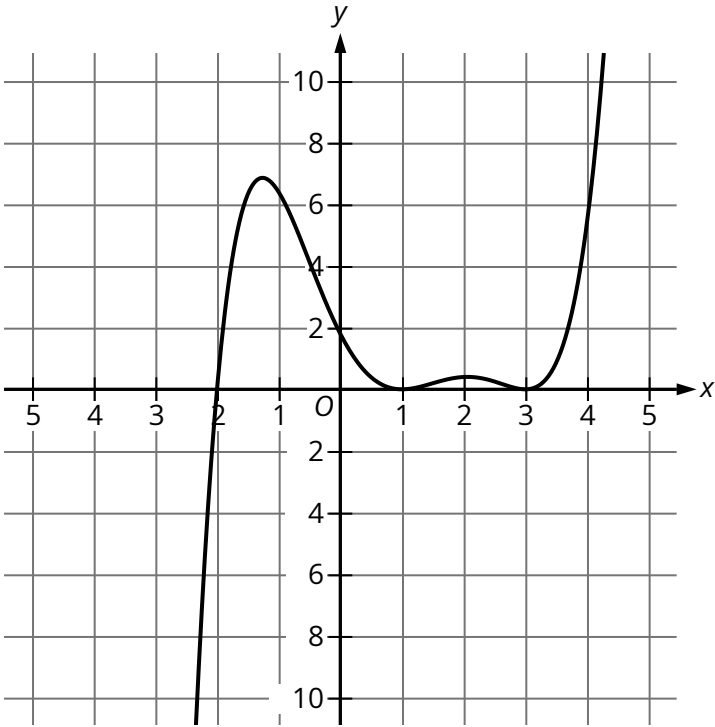


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question



The graph of $y = g(x)$ is shown. If $g(x + 28) = \frac{1}{10}(x - a)^2(x - b)^2(x - c)$, where a , b , and c are constants, which of the following could be the value of b ?

- Answer
- A. 31
 - B. 3
 - C. -2
 - D. -25

Correct Answer: D

Rationale

Choice D is correct. Substituting b for x in the given equation yields $g(b + 28) = \frac{1}{10}(b - a)^2(b - b)^2(b - c)$, which is equivalent to $g(b + 28) = \frac{1}{10}(b - a)^2(0)^2(b - c)$, or $g(b + 28) = 0$. It follows that when $x = b + 28$, $g(x) = 0$. There are three points where the graph of $y = g(x)$ intersects the x -axis: $(-2, 0)$, $(1, 0)$, and $(3, 0)$. Therefore, $g(-2) = 0$, $g(1) = 0$, and $g(3) = 0$. It follows that either $b + 28 = -2$, $b + 28 = 1$, or $b + 28 = 3$. Subtracting 28 from each side of the equation $b + 28 = -2$ yields $b = -30$. Subtracting 28 from each side of the equation $b + 28 = 1$ yields $b = -27$. Subtracting 28 from each side of the equation $b + 28 = 3$ yields $b = -25$. Thus, the value of b must be either -30 , -27 , or -25 . Of these values, only -25 is given as a choice.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This would be a possible value of b if the graph shown was the graph of $y = g(x + 28)$, not $y = g(x)$.

Choice C is incorrect. This would be the value of c , not the value of b , if the graph shown was the graph of $y = g(x + 28)$, not $y = g(x)$.